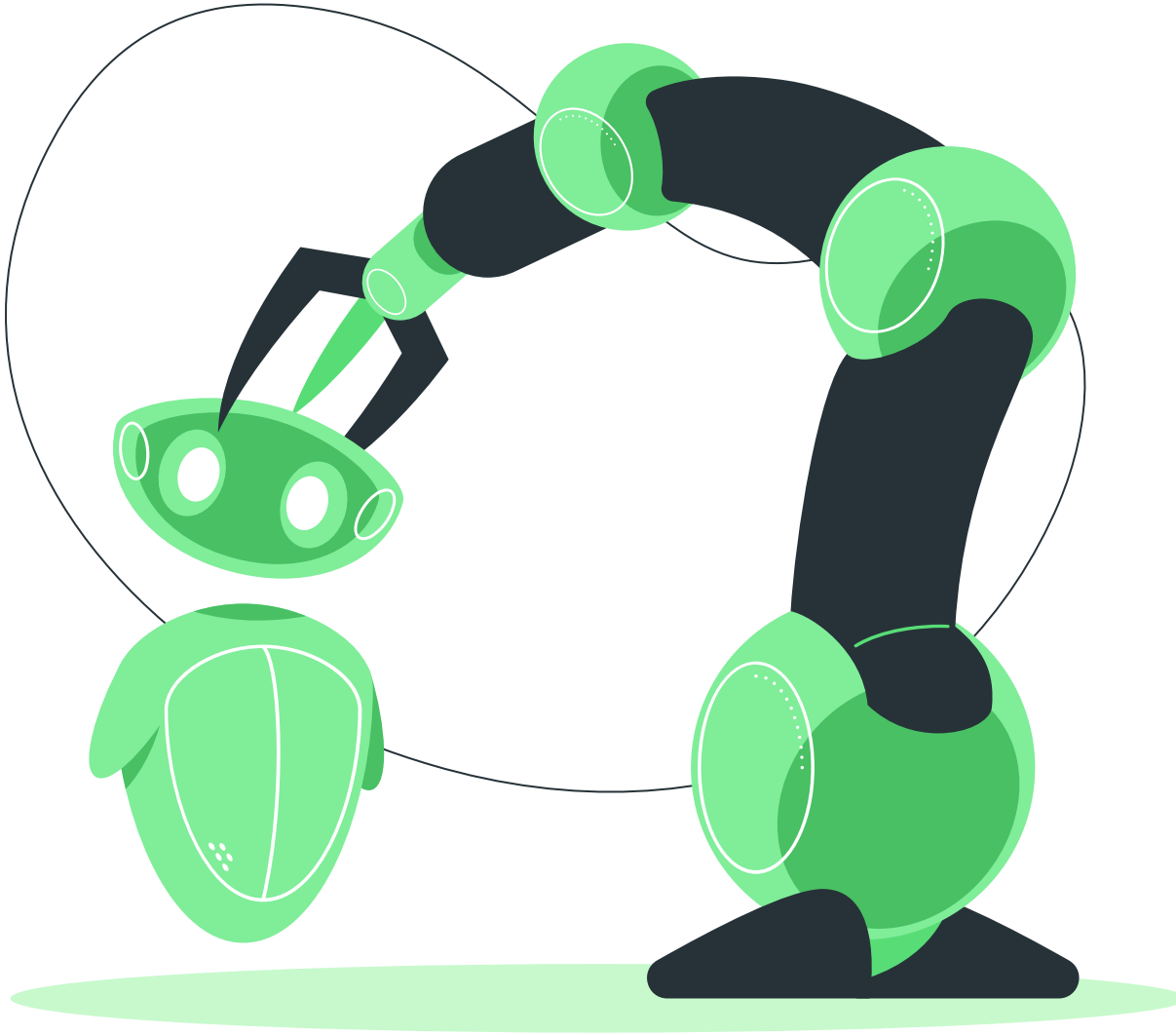
# Grp13: Traffic Officer in Disguise



# Table of Contents


1 **Introduction**
Origin of idea

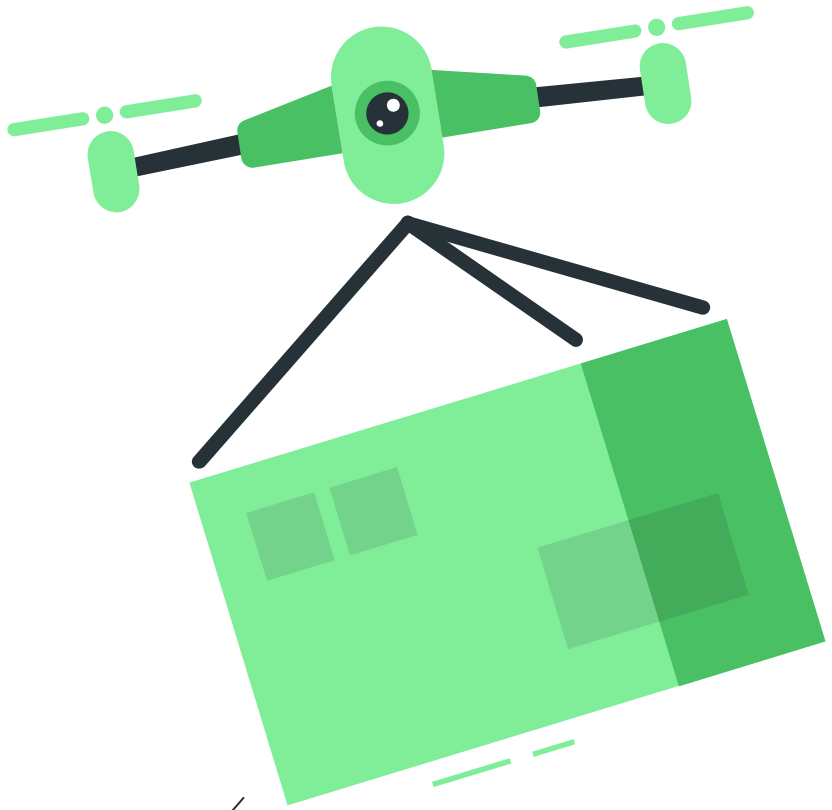
2 **Methodology and Description**
->Milestone Achievement
->Implementation

3 **Result and Evaluation**
Evaluation of the project results

4 **Conclusion**
Thoughts and remarks for future development

# 1 Introduction

Origin of Idea



# Introduction

Observations from daily life:

- daily routine of going back school
- common things: Street lamps

'What if it has a secret function?'



# Introduction




2 Implemented Modules:

- Traffic Road Status observer (Traffic Officer)

- Energy-saving Street Lamp ('bush' to disguise the traffic officer)

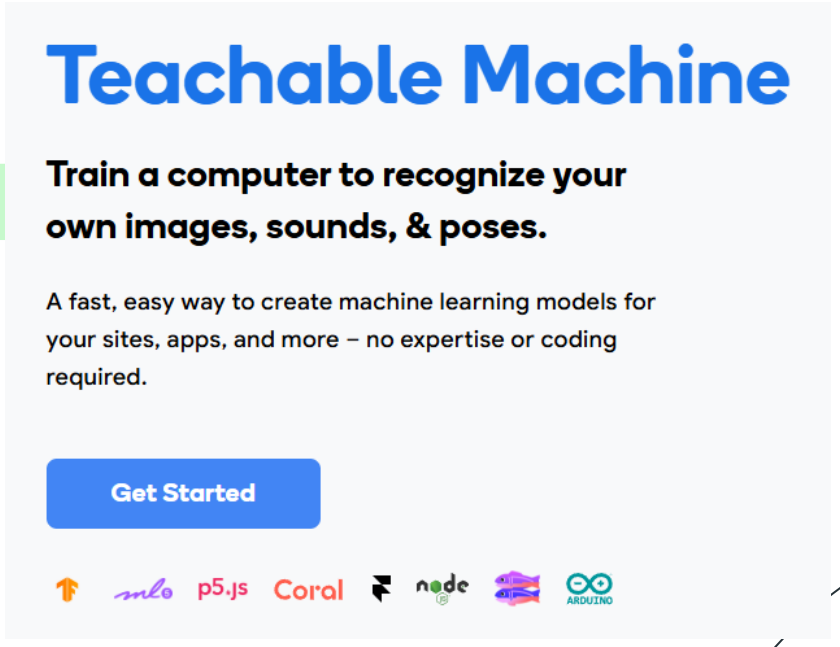


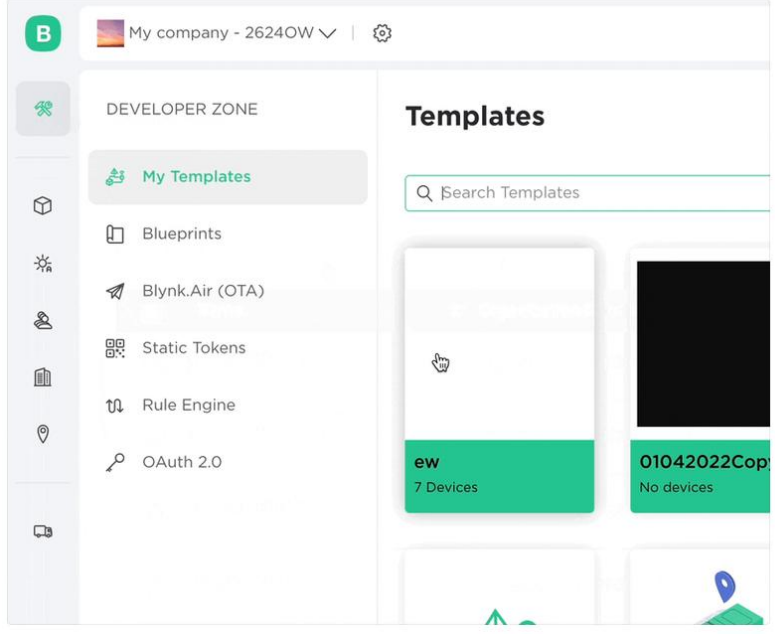
**-->Project "Traffic Officer in Disguise"**

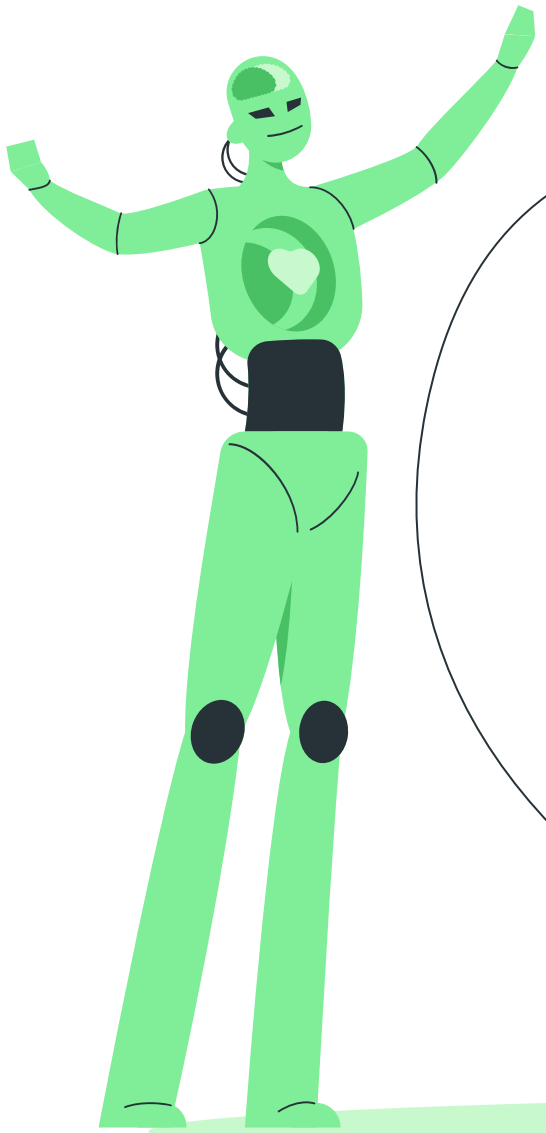
# Introduction

Software tools/ platforms used

- Teachable machine
- Blynk Template
- Light varying output based on light sensor





# 2 Methodology and Description

(Including Milestone and Implementation)



# Traffic status Module And Street lamp module

**Hardware**

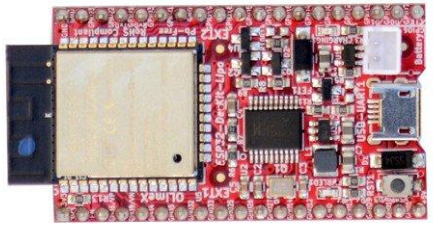
-nano rp2040
-ESP32
-Camera
-RGB LED
-buzzer

**Software**

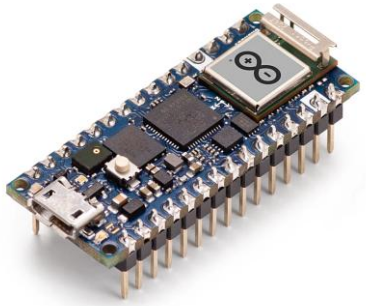
-Techable Machine
-Blynk Template
-Arduino IDE/ Cloud Editor
-VS code









# Traffic status module (Techable Machine)

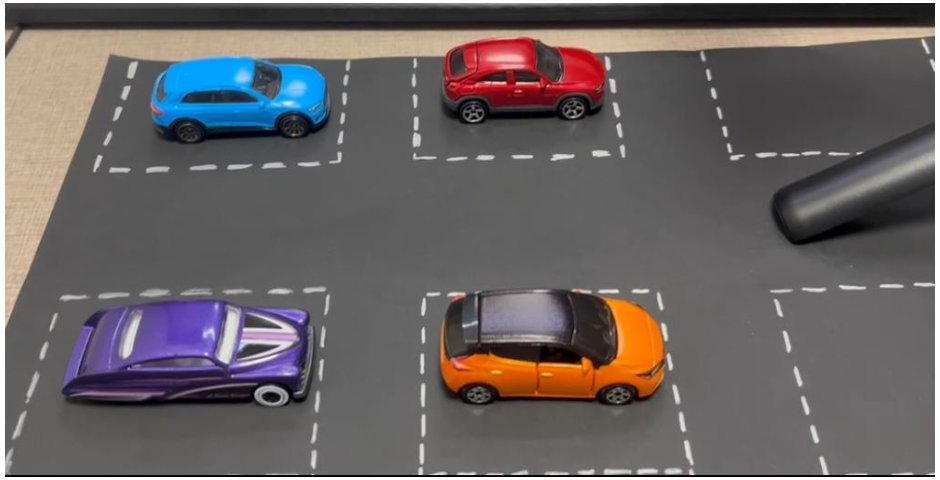
## 4 labels for image recognition of parking

| | |
|---|---|
| 1% | Fully Parked |
| 2% | Parked not well |
| 85% | Parking available |
| 50% | Anomaly |

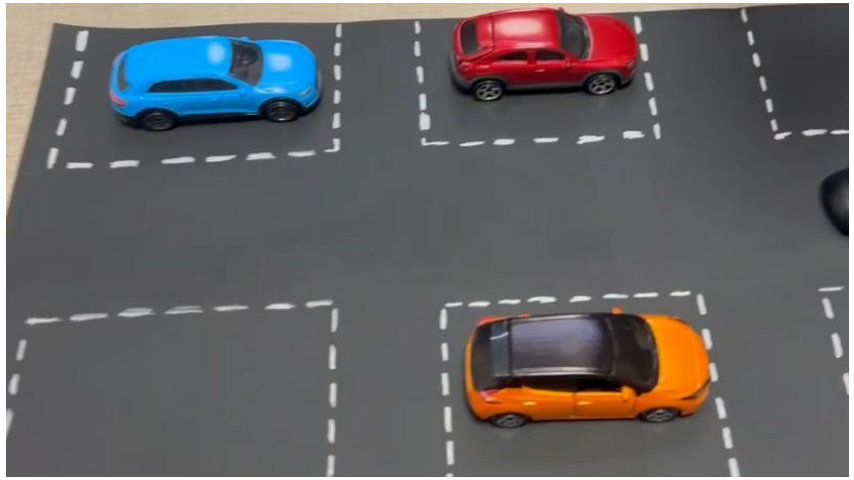


# 4 labels for image recognition of parking

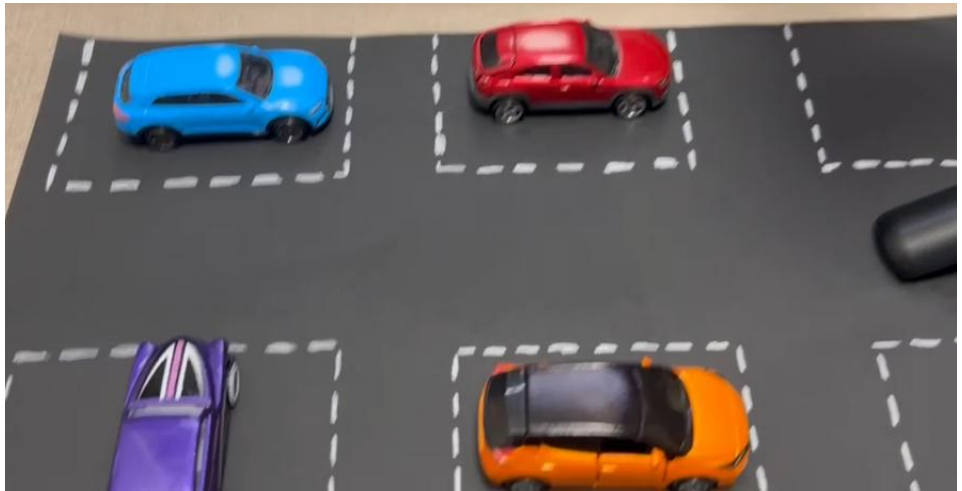
**Fully parked -> RED LED**



**Parking Available->GREEN LED**



**Parked not Well -> YELLOW LED**



**Anomaly->Buzzer on**

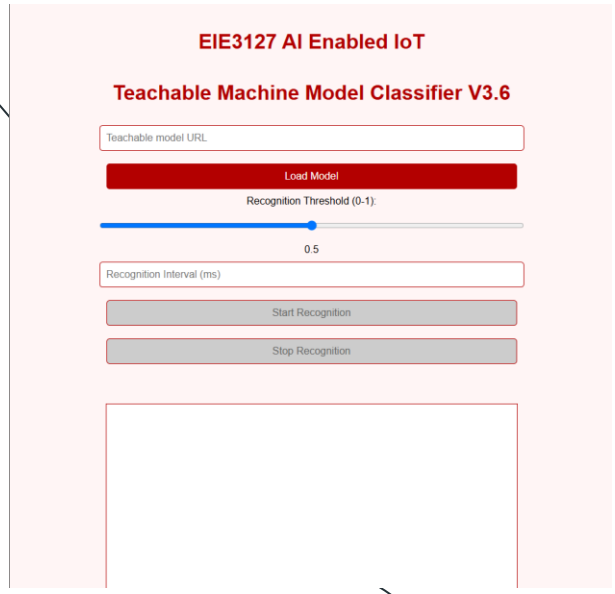


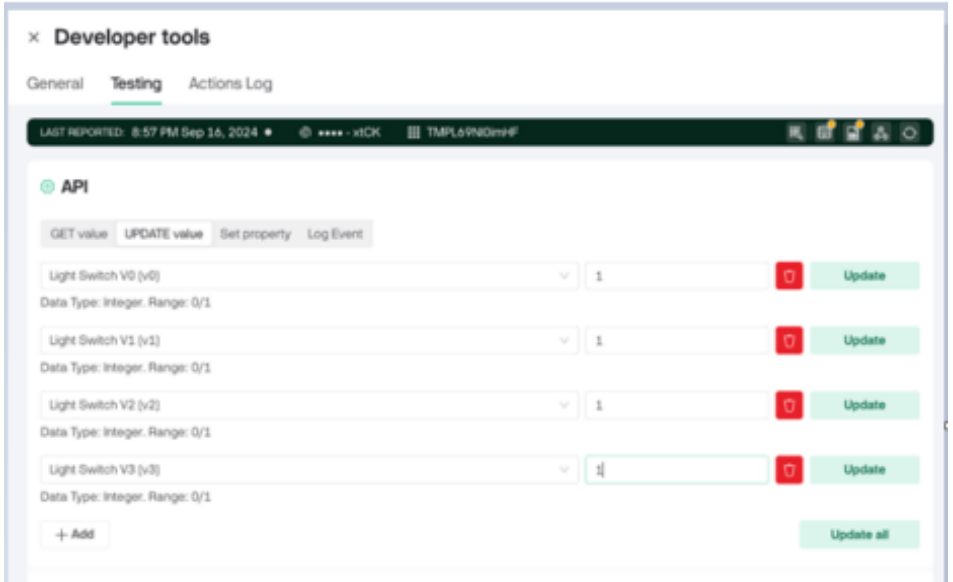
# Teachable machine with Blynk

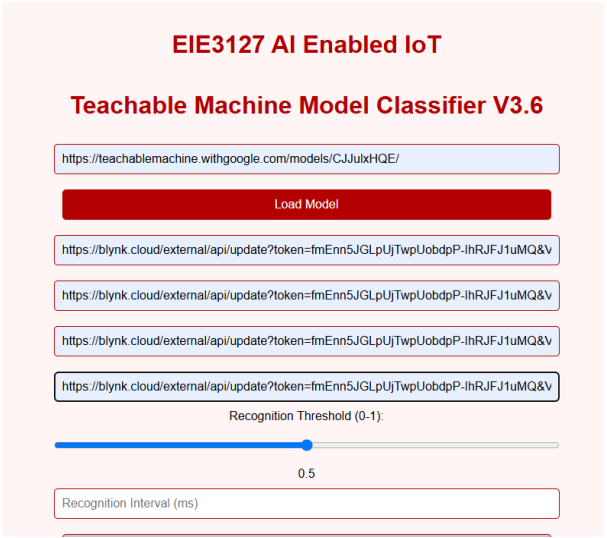
## PolyU's EIE Teachable Machine model Classifier【1】

## Blynk platform using 4 API links

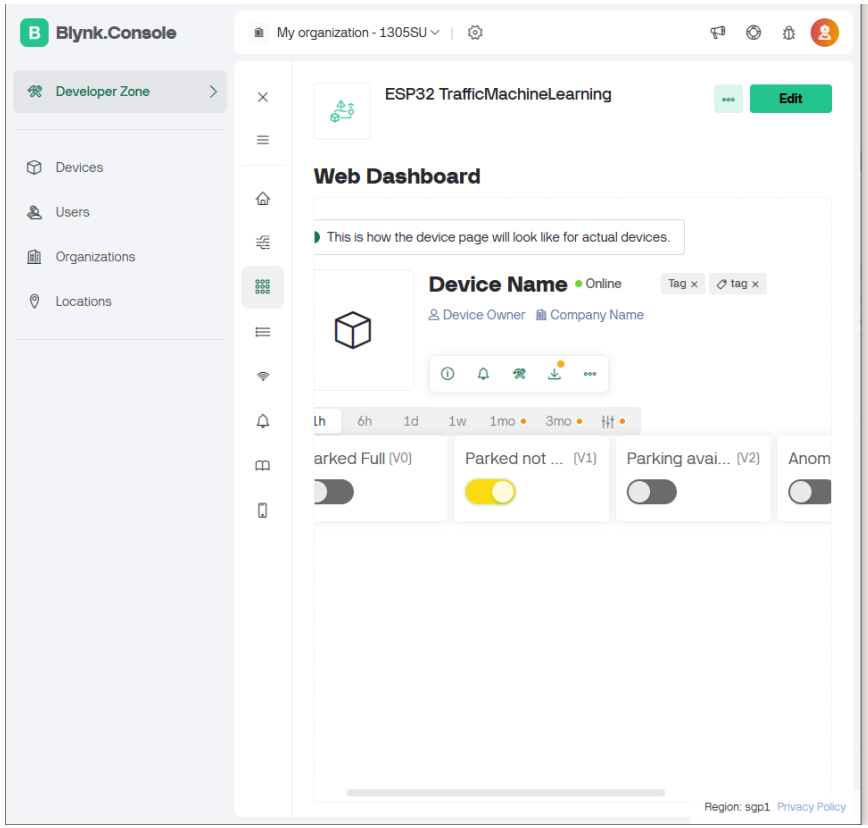
## Integrating Teachable machine model and Blynk platform

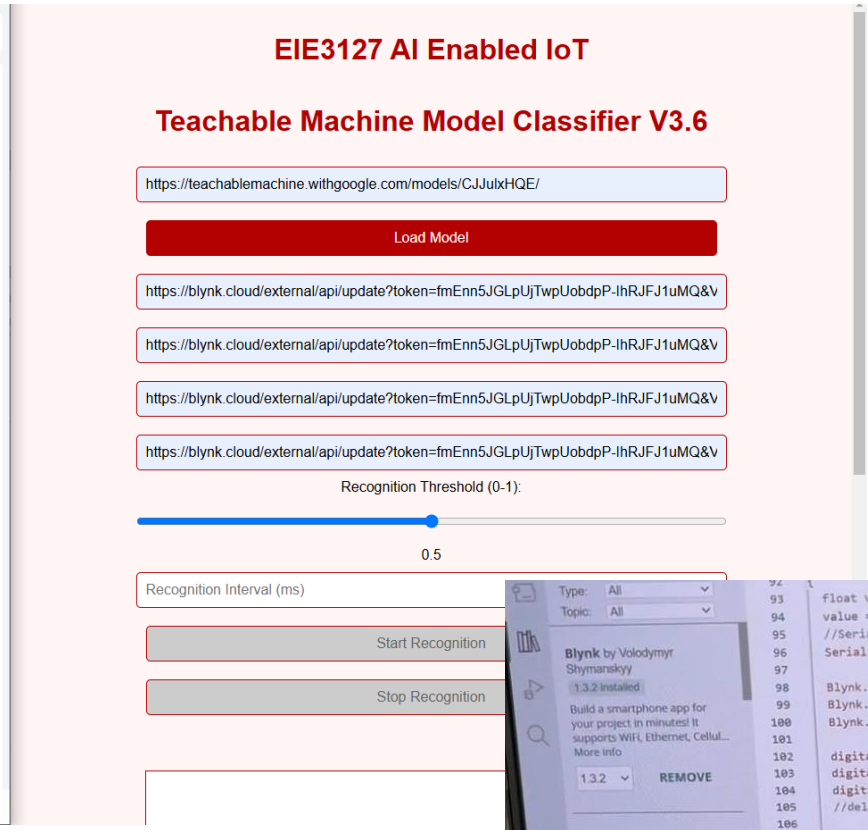


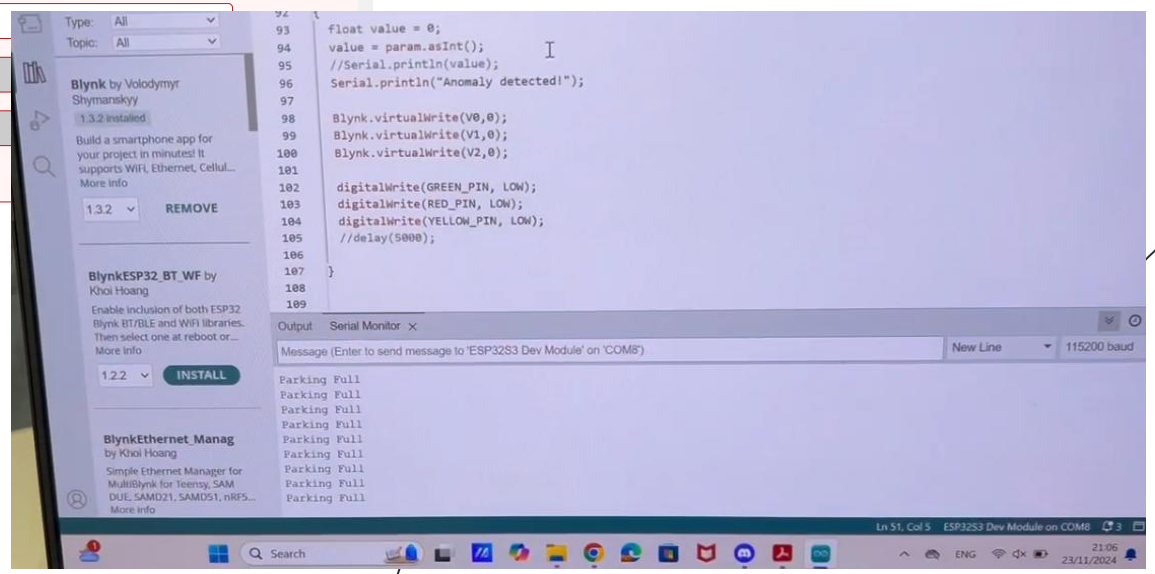



# Traffic Status module (Setup finish)



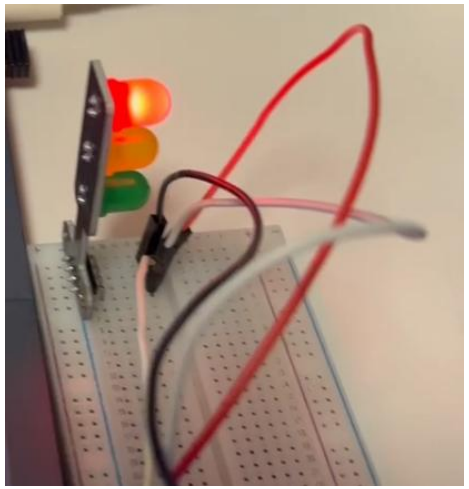



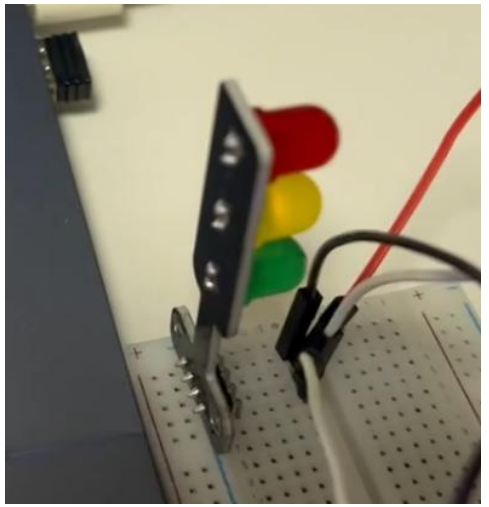



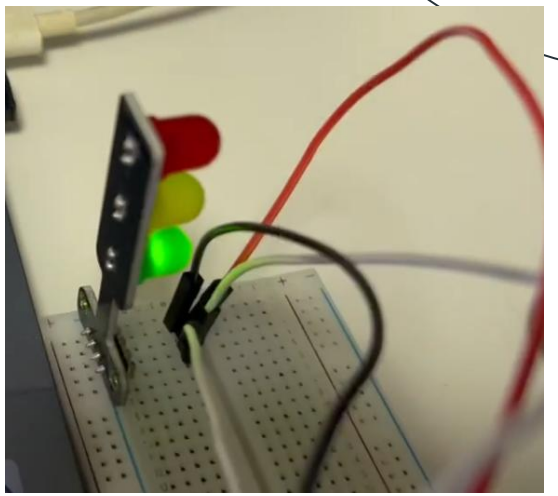

(With ESP32S3 coding sketch and hardware setup ready on the side)

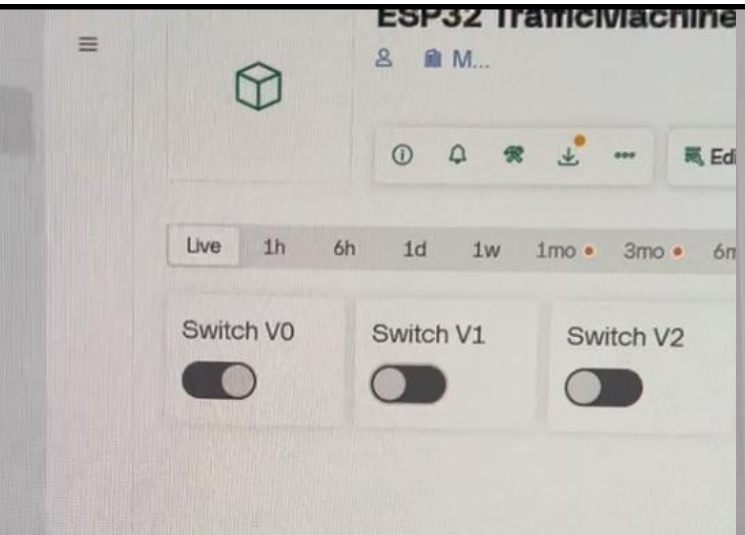
# Milestone and achievements

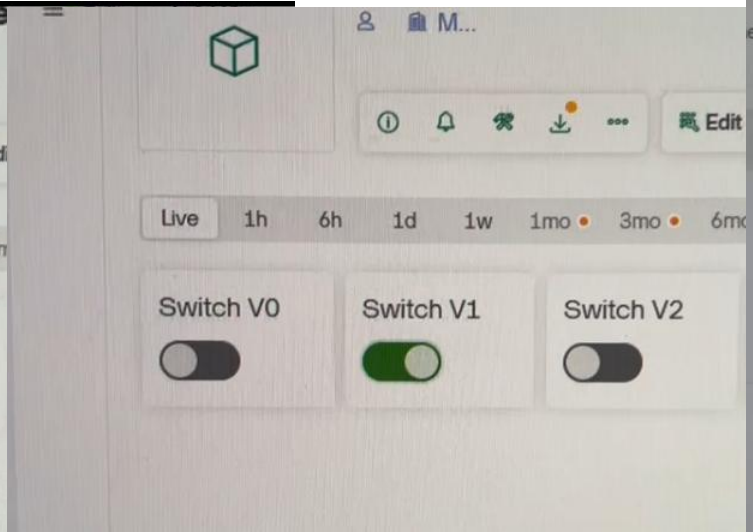
**Consistent accuracy of identifying labels and synchronization of Blynk Platform**

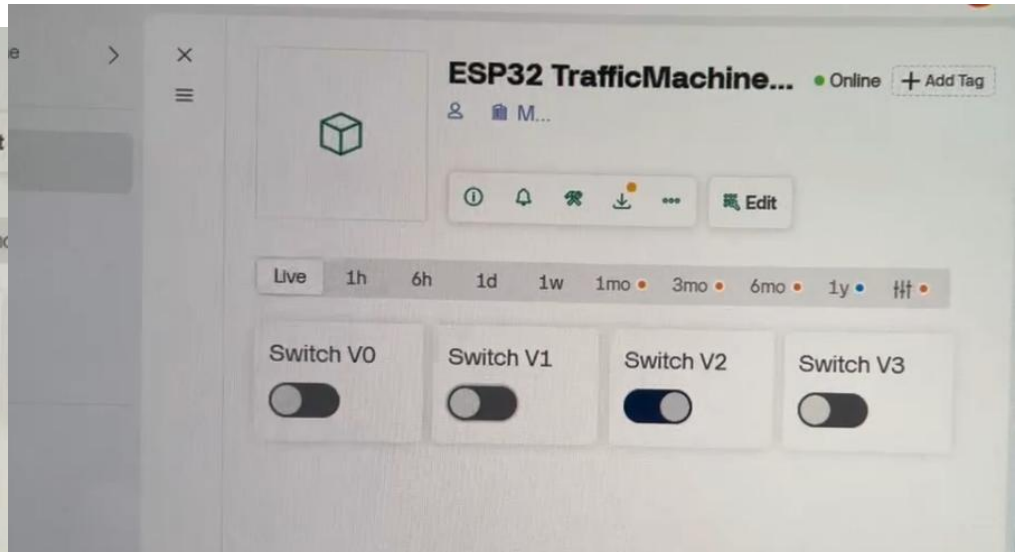








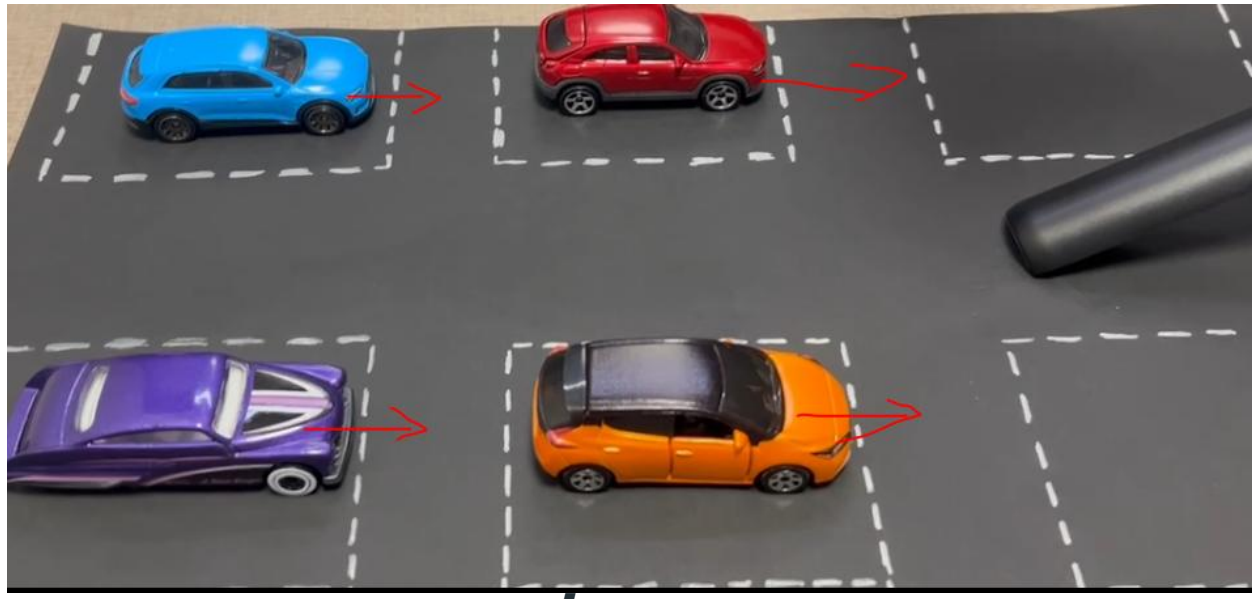
# Challenges faced

-Controlled variable required for better accuracy

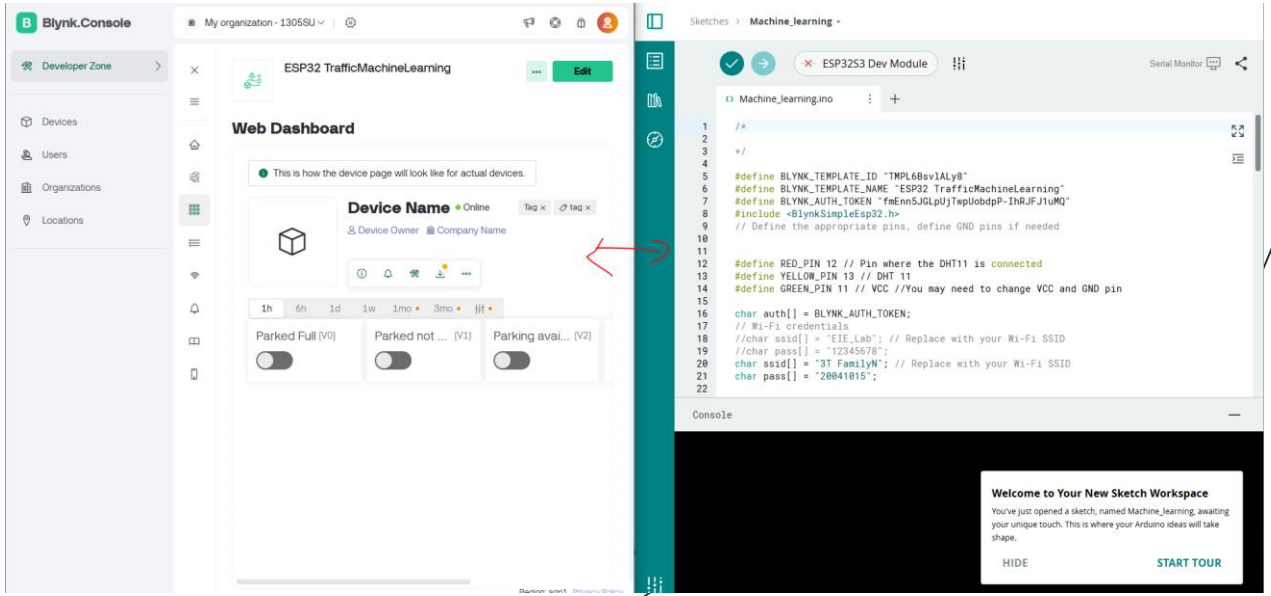
-compatibility of cloud editor and Blynk Platform




## Cars faced to same direction (controlled variable)



## Unstable connection/synchronization to give serial outputs on the serial monitor



# Implementation flow

**1. Teachable machine**

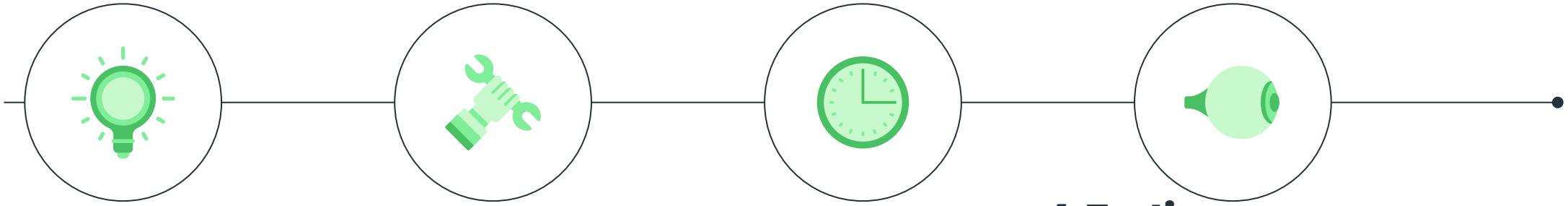
AI model setup

**2. Blynk template setup**

Virtual pins and API link setup

**3. Arduino IDE/Cloud Editor**

-Program code setup of connecting Blynk API and ESP32
-LED and buzzer hardware setup



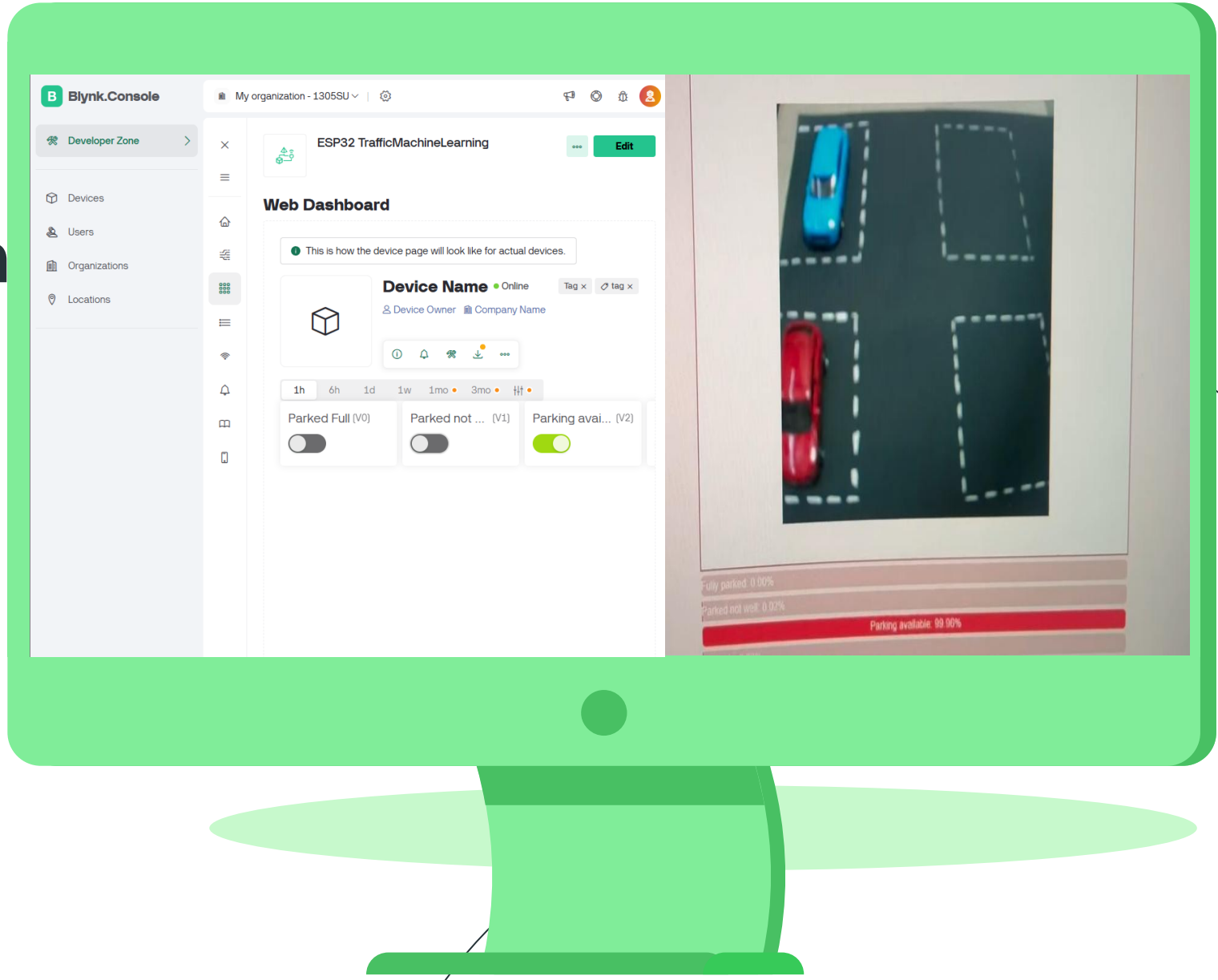
**4. Testing connectivity between the Blynk template and classifier website**

Putting Blynk API links and exported AI model to Classifier Website

# Results and Evaluation

-showing fairly accurate results

-successful integration of hardware outputs and software inputs of label



# Conclusion and Remarks



-successful yet not practical

-More sampling of different car subjects are required to make it practical (ex: people crossing streets, car types)





# Thank you for watching.

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